

EXAMPLE CLAIM-STYLE STATEMENTS (NON-LIMITING) AND SCOPE CLARIFICATIONS

A. Overview

The following statements are provided as non-limiting examples of claim-style scope for a provisional filing. These statements are illustrative only and are intended to describe combinations of elements disclosed herein, including multiple nodes, a computing system, privacy-preserving outputs, cross-device correlation, and cross-user correlation under consent gating.

B. Example System Statements

1. A system comprising: (i) a plurality of sensing nodes configured to collect intimate physiology signals and/or geometry signals, and (ii) a computing system configured to receive the signals, generate privacy-preserving metrics, and present outputs without explicit anatomical imagery.
2. The system of statement 1, wherein the plurality of sensing nodes includes at least one external wearable node configured to measure tumescence- and rigidity-related behavior over time using one or more of pressure sensing, strain sensing, magnetic sensing, optical pulse sensing, and inertial sensing.
3. The system of statement 1, wherein the plurality of sensing nodes includes at least one intraluminal wearable node configured to measure longitudinal physiology over time using one or more of impedance sensing, multi-frequency impedance sensing, capacitive wetness/dielectric sensing, pH sensing, temperature sensing, distributed pressure sensing, and inertial sensing.
4. The system of statement 1, wherein the plurality of sensing nodes includes at least one session-based external geometry node configured to measure geometry using one or more of optical flow sensing, time-of-flight sensing, lidar sensing, infrared sensing, inertial sensing, and contact confirmation.
5. The system of statement 1, wherein the plurality of sensing nodes includes at least one intraluminal geometry/compliance node configured to measure internal distance and compliance using endpoint sensing and controlled radial stimulus including an inflatable cuff or microbladder ring, and further configured to measure stimulus pressure using a pressure transducer and contact distribution using a circumferential sensing element.
6. The system of statement 1, wherein the plurality of sensing nodes includes at least one upper-body context node configured to measure one or more of ECG, PPG, inertial sensing, respiratory proxies, temperature context, electrodermal activity, and wear integrity.

C. Example Method Statements

7. A method comprising: receiving data from two or more sensing nodes; conditioning the data using signal quality indices and artifact rejection; generating privacy-preserving metrics; and performing cross-device correlation between metrics from different nodes to produce correlation outputs comprising an alignment score, a lag signature, and/or a confidence indicator.
8. The method of statement 7, further comprising conditioning interpretation of intimate-region metrics based on context state derived from an upper-body context node, including sleep proxies, activity state, autonomic proxies, thermal context, and/or wear-integrity state.

9. The method of statement 7, wherein the cross-device correlation is computed within context-tagged time windows comprising low-motion windows, sleep windows, rest windows, or other validated windows.

D. Cross-User Correlation Under Consent

10. A method comprising: obtaining authorization from a first user and a second user; generating privacy-preserving metrics for each user from one or more sensing nodes; and computing cross-user correlation using the privacy-preserving metrics to output one or more of an alignment score, timing synchrony score, stability score, and confidence indicator, without sharing raw signals between the users.

11. The method of statement 10, wherein the cross-user correlation supports pairing contexts including male-to-female, male-to-male, female-to-female, and other pairings, while maintaining privacy-preserving output formats without explicit anatomical imagery.

E. Non-Limiting Scope Clarifications

12. Any node described herein may be implemented in multiple form factors and may include substituted sensor modalities while retaining the functional intent described herein.

13. Processing may be performed on-node, on a user device, on a remote server, or distributed across components, and may include rules-based processing and/or machine learning.

14. Outputs may be presented as abstract indices, zones, ranges, and curves, and may include confidence indicators and data-quality labeling.

15. Any workflow described herein may be performed with any subset of nodes and in any order, including baseline-first workflows and longitudinal-first workflows.